

Monsoon Semester Thesis Report
On
Implementation and Defense Against Adversarial
Attacks in Image Classification Model



*Submitted to **Indian Institute of Technology (ISM) Dhanbad** in
partial fulfilment of the requirements for the award of the degree of*

Master of Technology

(Computer Science and Engineering)

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Introduction

Adversarial attacks exploit the vulnerabilities of machine learning models, particularly deep learning models like CNNs, by introducing carefully crafted perturbations to the input data. These perturbations are usually small enough to be imperceptible to the human eye but can cause the model to misclassify the input. For example, an image of a panda might be subtly altered so that a CNN model mistakenly identifies it as a gibbon, while to a human observer, the image still clearly appears to be a panda.

The primary goal of these attacks is to fool the model into making incorrect predictions, either by changing the model's output to a specific incorrect label (targeted attack) or simply causing any incorrect prediction (untargeted attack). This can have serious implications, especially in sensitive applications like autonomous driving, medical diagnosis, or security surveillance.

CNN models learn to recognize patterns in images by using a series of convolutional filters and non-linear transformations. However, they can be overly sensitive to small, seemingly random changes in the input data. Adversarial attacks exploit this by adding a calculated "noise" to the input image that shifts the model's decision boundary.

Problem Statement

The rapid advancement of deep learning models, particularly in image classification, has led to their widespread deployment in critical applications, such as autonomous vehicles, healthcare diagnostics, and security surveillance.

However, these models are vulnerable to adversarial attacks, where small, imperceptible perturbations to input images can cause significant misclassifications.

This project aims to explore the effectiveness of adversarial attacks, specifically the Fast Gradient Sign Method (FGSM), on a deep learning-based image classification model

Research Gap

Limitations of Existing Defensive Techniques:

Several defensive methods have been explored, such as dropout, input transformation techniques, feature squeezing, and model ensembles. While dropout is effective in preventing overfitting, its primary purpose is not to enhance adversarial robustness, limiting its effectiveness against targeted adversarial attacks. Input transformation techniques, including Gaussian noise addition, image compression, and resizing, are designed to obscure adversarial noise. However,

these methods often degrade the quality of the input data, leading to reduced model accuracy on clean (non-adversarial) examples. Similarly, feature squeezing, which reduces the sensitivity of models to input variations, can inadvertently lower the model's ability to accurately classify genuine data, thus impacting its overall performance.

Effectiveness Against Advanced Attacks:

Despite the promise shown by existing defenses, they are largely ineffective against sophisticated adversarial attacks. Techniques like feature squeezing may work against simple noise-based attacks but fail against gradient-based approaches like the Carlini & Wagner (C&W) attack or Projected Gradient Descent (PGD). Additionally, model ensembles, which rely on the consensus of multiple models to improve robustness, can be computationally expensive and may not provide a significant advantage against highly adaptive attacks. This indicates a lack of comprehensive defenses that can generalize well across various attack types.

Need for More Robust Defense Mechanisms:

Given the identified gaps in existing solutions, there is a pressing need to explore and develop more robust methods to defend against adversarial attacks. Current defenses either lack comprehensive effectiveness against sophisticated attacks or impose high computational demands, making them unsuitable for widespread adoption. This research aims to address these gaps by investigating the Fast Gradient Sign

Method (FGSM) and exploring the potential of adversarial training as a more effective defense mechanism. By focusing on these areas, the goal is to contribute to the development of more resilient deep learning models that can withstand a wider range of adversarial threats.

Thesis Objective

This research focuses on understanding and addressing the vulnerabilities of deep learning models, particularly Convolutional Neural Networks (CNNs), to adversarial attacks. The study is driven by the following key objectives:

1. Implement the Fast Gradient Sign Method (FGSM) to Generate Adversarial Examples:

The first goal of this research is to implement the Fast Gradient Sign Method (FGSM), a widely recognized technique for generating adversarial examples. FGSM is a gradient-based attack method that perturbs the input data by adding a small, calculated noise in the direction of the gradient of the loss function. enough to remain almost The perturbation is designed to maximize the model's prediction error while being subtle imperceptible to the human eye. By implementing FGSM, the research aims to generate adversarial examples that can be used to evaluate the vulnerability of CNN models. This implementation will serve as a foundation for assessing how such perturbations impact

model predictions, helping to highlight the weaknesses in current deep learning systems.

$$x^{adv} = x + \epsilon \cdot \text{sign}(\nabla_x J(x, y_{true})),$$

where

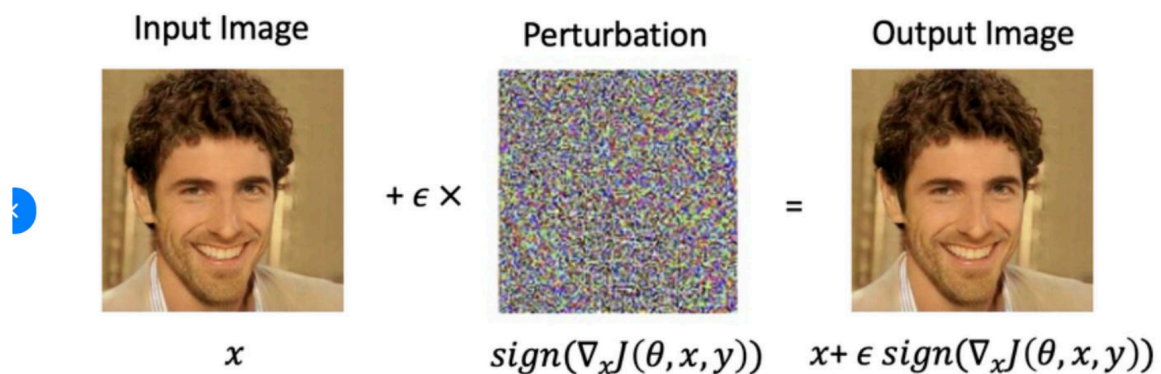
x is the input (clean) image,

x^{adv} is the perturbed adversarial image,

J is the classification loss function,

y_{true} is true label for the input x .

For example :



2. Evaluate the Impact of Adversarial Attacks on Model Performance:

The second objective is to assess the impact of adversarial attacks on the performance of CNN models. This involves testing the model on both clean and adversarially perturbed datasets to measure the drop in accuracy and other performance metrics. The evaluation will focus on understanding how adversarial examples affect model confidence and classification accuracy. By analyzing metrics such as misclassification rates, confusion matrices, and confidence scores, the research seeks to quantify the extent to which adversarial attacks can degrade the performance of CNNs. The goal is to provide insights into the specific vulnerabilities that adversarial attacks exploit, which can guide the development of more effective defense mechanisms.

3. Explore Defensive Strategies, Such as Adversarial Training, to Improve Model Robustness:

The final key goal is to investigate defensive techniques aimed at enhancing the robustness of CNN models against adversarial attacks. Among these strategies, **adversarial training** is a primary focus. Adversarial training involves augmenting the training dataset with adversarial examples, allowing the model to learn from both clean and adversarially perturbed inputs. This technique aims to make the model more

resilient by forcing it to develop features that are less sensitive to small input perturbations.

The research will explore different configurations of adversarial training, such as varying the magnitude of perturbations adjusting the ratio of adversarial examples in the training set, and experimenting with different loss functions to enhance robustness. Additionally, the study will compare the effectiveness of adversarial training with other defense mechanisms, such as dropout, input transformation techniques, and model ensembles, to provide a comprehensive evaluation of their strengths and weaknesses.

By exploring these defensive strategies, the research aims to identify the most effective methods for mitigating the impact of adversarial attacks. The ultimate goal is to contribute to the development of deep learning models that are not only accurate but also robust against adversarial threats, thereby improving their reliability in real-world applications.

Summary of Key Goals

To summarize, this research aims to:

- **Generate adversarial examples using FGSM** to understand how small perturbations can mislead deep learning models.
- **Evaluate the impact of these attacks on model performance**, highlighting the vulnerabilities of CNNs to adversarial inputs.
- **Investigate defensive techniques like adversarial training** to enhance model robustness, aiming to build more resilient models that can withstand adversarial threats.

By achieving these goals, the research intends to advance the field of adversarial machine learning, providing insights into both the attack strategies and the defense mechanisms that can protect deep learning models from malicious exploitation.

Experimental Results

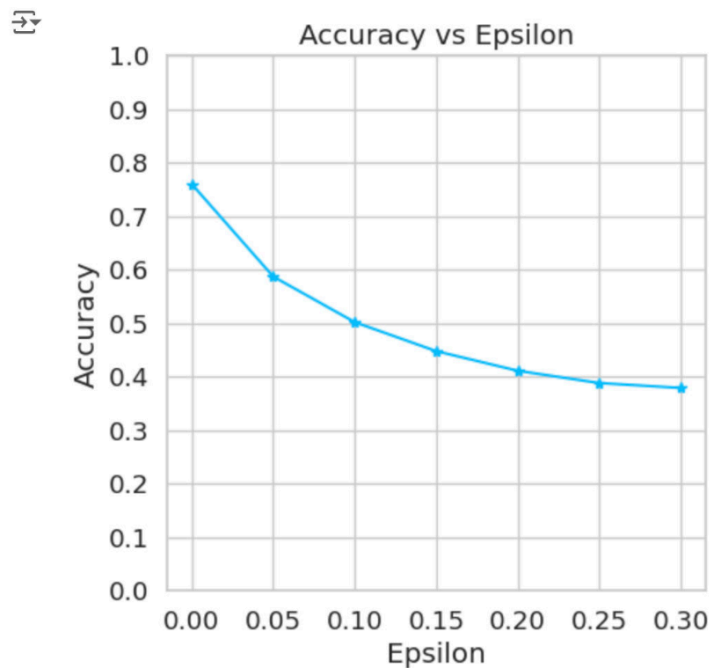
In this research, we leverage a pre-trained ResNet (Residual Network) architecture for the task of image classification using PyTorch. ResNet is a widely acclaimed deep learning model known for its effectiveness in image recognition tasks due to its use of residual connections, which help mitigate the vanishing gradient problem in deep neural networks. The ResNet model, originally trained on the ImageNet dataset with over 1 million images spanning 1000 different classes, has already learned a rich set of features such as edges, textures, and shapes, which are transferable to various other image classification tasks.

To adapt this pre-trained ResNet model for our specific use case, we employ transfer learning—a technique that allows us to take advantage of the knowledge gained by the model from its extensive training on the ImageNet dataset. Instead of training a model from scratch, which would require a large dataset and significant computational resources, transfer learning enables us to fine-tune the existing model for our specific dataset.

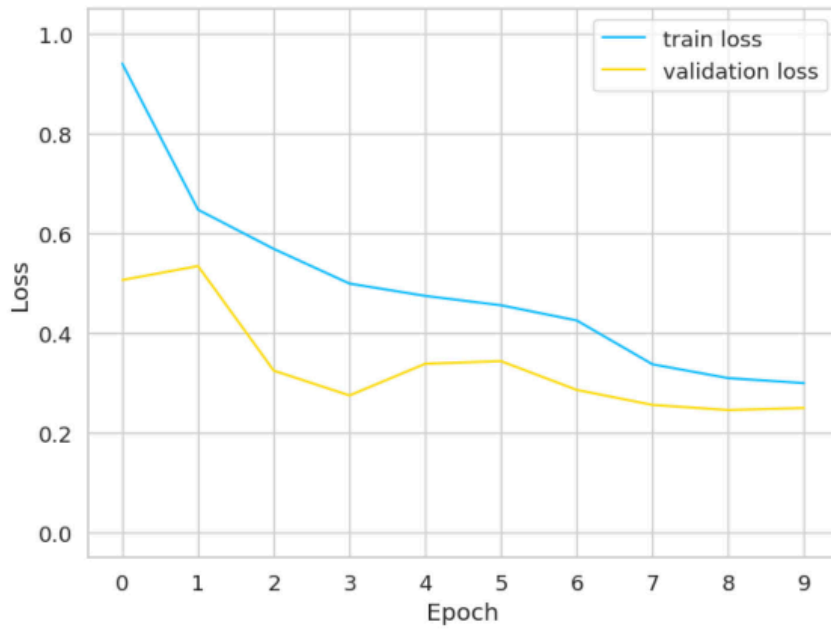
Since our dataset consists of images belonging to 6 distinct classes, we need to modify the final layer of the ResNet model. The original architecture is designed to output predictions across 1000 classes, using a fully connected layer with 1000 output neurons. To tailor the model for our classification task, we adjust this final layer to have only 6 output neurons, each corresponding to one of the classes in our dataset. This adjustment involves replacing the last fully connected layer with a new one that matches the number of target classes.

By freezing the earlier layers of the model and only fine-tuning the weights of the newly added layer, we can efficiently train the model with a smaller dataset while still benefiting from the robust feature extraction capabilities of the pre-trained ResNet. This approach significantly reduces training time and improves model performance, as the earlier layers retain the ability to recognize generic patterns and features, while the last layer learns to classify our specific set of classes. Thus, transfer learning with ResNet provides a powerful and efficient solution for image classification tasks with limited data availability.

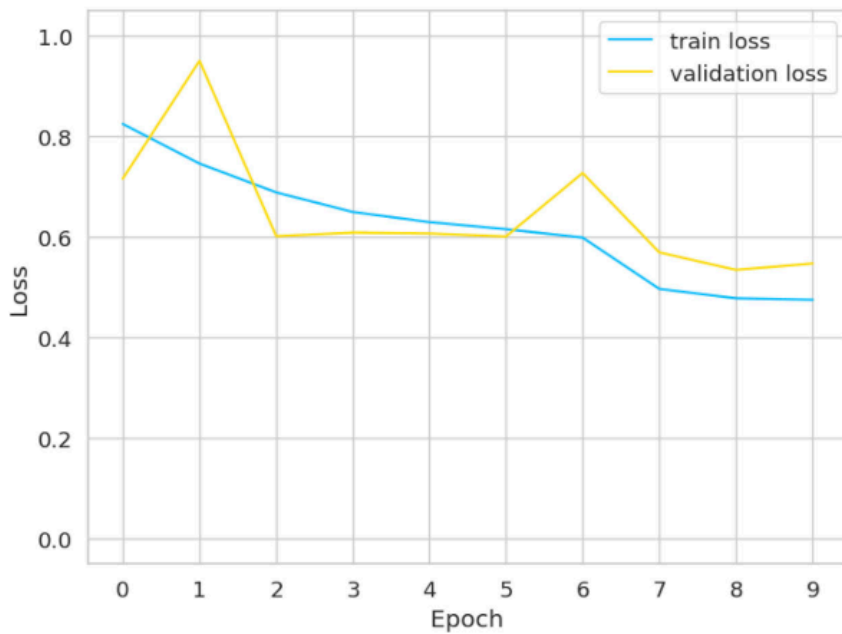
Accuracy vs Epsilon



The impact of epsilon ϵ on model accuracy in FGSM attacks highlights the trade-off between attack effectiveness and perceptibility. Understanding this relationship is crucial for designing robust deep learning models capable of resisting adversarial attacks. By carefully tuning epsilon ϵ and employing defensive strategies like adversarial training, it is possible to enhance model security against adversarial threats.



Before Attack



After Attack

Testing
on

Evaluating Model Robustness on Unmodified Images

Testing on unmodified images serves as a baseline to understand the model's original classification capabilities. A robust model should not only defend against adversarial examples but also maintain high accuracy on clean images. If a model is overly tuned to defend against adversarial attacks, it may become less effective on normal, real-world data, leading to a phenomenon known as **accuracy trade-off**. This trade-off happens when the techniques used to harden the model against adversarial inputs inadvertently reduce its performance on non-adversarial, clean inputs.

	precision	recall	f1-score	support
buildings	0.92	0.92	0.92	349
forest	0.99	0.99	0.99	379
glacier	0.88	0.88	0.88	442
mountain	0.89	0.87	0.88	420
sea	0.93	0.96	0.95	408
street	0.93	0.94	0.93	400
accuracy			0.92	2398
macro avg	0.92	0.93	0.93	2398
weighted avg	0.92	0.92	0.92	2398

```

For Epsilon =0
/usr/local/lib/python3.10/dist-packages/torch/utils/data/dataloader.py:169:
  warnings.warn(_create_warning_msg(
precision    recall  f1-score   support

 buildings    0.67    0.91    0.77     349
  forest      0.99    0.74    0.85     379
  glacier     0.70    0.82    0.76     442
 mountain     0.86    0.70    0.77     420
    sea       0.65    0.87    0.74     408
  street      0.89    0.54    0.67     400

 accuracy                0.76     2398
 macro avg              0.79    0.76    0.76     2398
 weighted avg           0.79    0.76    0.76     2398

```

Epsilon = 0

```

For Epsilon =0.1
/usr/local/lib/python3.10/dist-packages/torch/utils/data/dataloader.py:169:
  warnings.warn(_create_warning_msg(
precision    recall  f1-score   support

 buildings    0.49    0.79    0.60     349
  forest      0.91    0.60    0.72     379
  glacier     0.40    0.51    0.45     442
 mountain     0.55    0.41    0.47     420
    sea       0.40    0.55    0.46     408
  street      0.57    0.20    0.30     400

 accuracy                0.50     2398
 macro avg              0.55    0.51    0.50     2398
 weighted avg           0.55    0.50    0.49     2398

```

Epsilon = 0.1

After Adversarial Attack Training

After training a deep learning model with adversarial examples (known as adversarial training), it is critical to evaluate the model's performance not only on adversarially perturbed images but also on unmodified (clean) images. The purpose of this evaluation is to ensure that adversarial training does not significantly degrade the model's accuracy on real-world, unmodified data.

	precision	recall	f1-score	support
buildings	0.76	0.84	0.80	349
forest	0.95	0.89	0.92	379
glacier	0.84	0.70	0.76	442
mountain	0.82	0.72	0.76	420
sea	0.70	0.89	0.79	408
street	0.82	0.82	0.82	400
accuracy			0.81	2398
macro avg	0.81	0.81	0.81	2398
weighted avg	0.81	0.81	0.81	2398

Conclusions

Vulnerability of Deep Learning Models: The project demonstrated that deep learning models, even those with robust architectures like ResNet, are highly vulnerable to adversarial attacks.

Effectiveness of FGSM: Even small perturbations generated using the Fast Gradient Sign Method (FGSM) can have a significant impact on model performance, leading to a notable reduction in accuracy.

Sensitivity to Perturbations: The study highlighted how sensitive deep learning models are to subtle changes in input data, with the adversarial noise often being imperceptible to human observers but still capable of causing misclassifications.

Importance of Adversarial Defense: The results underscore the need for effective defensive strategies to protect models from adversarial perturbations, especially in critical applications like image classification.

Adversarial Training: While adversarial training can improve robustness to attacks, it also highlights the trade-off

between achieving robustness and maintaining high accuracy on clean (unmodified) data.

Model Robustness: The project emphasizes that while architectures like ResNet provide strong feature extraction capabilities, their susceptibility to adversarial attacks calls for a combination of strategies to enhance their robustness against these threats.

References

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